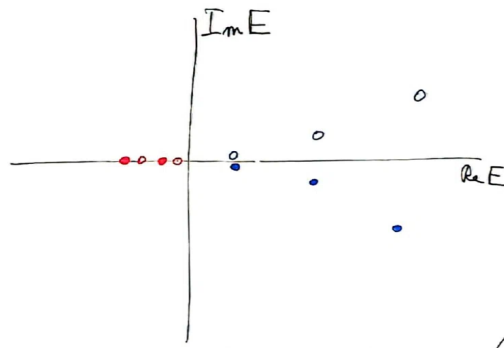
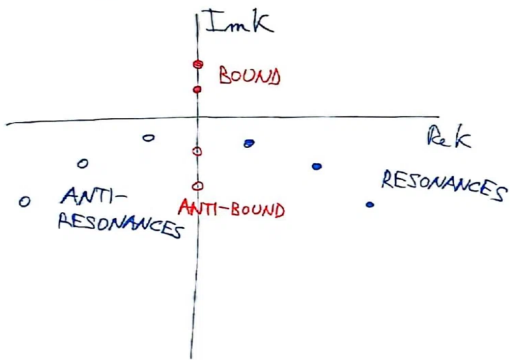
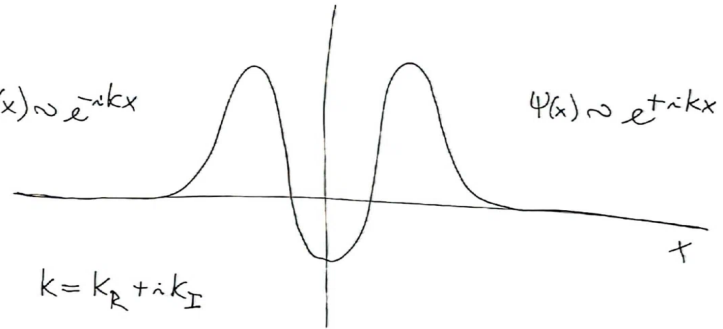


SIEGERT STATES



$$\psi(x) \sim e^{-ikx}$$

$$\psi(x) \sim e^{+ikx}$$



$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m} (k_R^2 - k_I^2) + 2i k_R k_I \frac{\hbar^2}{2m}$$

TISE

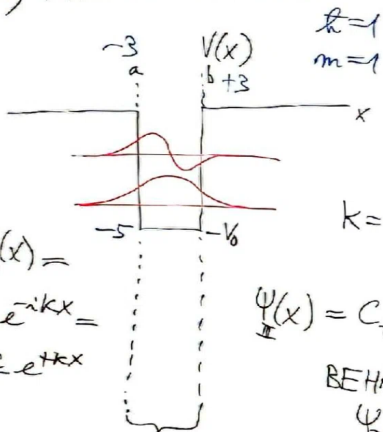
$$\left(-\frac{\hbar^2}{2m} \partial_{xx} + V(x) \right) \psi(x) = E \psi(x)$$

SIEGERT BOUNDARY CONDITIONS

$$\psi(x \rightarrow \pm\infty) = C_{\pm} e^{\pm ikx}$$

HOW TO SOLVE THE BOUNDARY VALUE PROBLEM (*)?

a) DIRECT METHOD



BOUND STATES

PICK UP TRIAL E
(OR EQUIVALENTLY TRIAL k)

$$k = ik, k > 0$$

$$\psi_I(x) = C_+ e^{+ikx} = C_+ e^{-kx}$$

BEHAVIOR AT THE BOUNDARY:

$$\psi_I(b) = \psi_{II}(b)$$

$$\psi_I'(b) = \psi_{II}'(b)$$

MIDDLE REGION: $a < x < b$

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) - V_0 \psi(x) = E \psi(x)$$

GENERAL SOLUTION

$$\psi''(x) = -\frac{2m}{\hbar^2} (E + V_0) \psi(x)$$

$$\psi(x) = A_+ e^{+ikx} + A_- e^{-ikx}$$

$$k = \sqrt{2m(E + V_0)} > 0$$

$$A_+ e^{+ikb} + A_- e^{-ikb} = C_+ e^{-kb}$$

$$A_+ e^{+ikb} - A_- e^{-ikb} = C_+ e^{-kb} (-k)$$

$$\psi_I(a) = \psi_{II}(a)$$

$$\psi_I'(a) = \psi_{II}'(a)$$

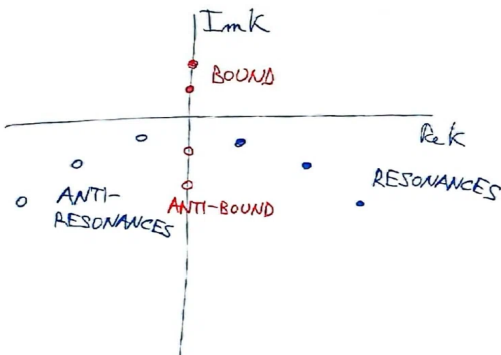
FIXING A_+, A_- UNIQUELY
(PROPORTIONAL TO C_-)

TRANSCENDENTAL
EQUATION FOR E

WHICH IS SATISFIED
ONLY FOR DISCRETE
BOUND STATE ENERGIES

BUT $C_- \sim C_+$
BECAUSE OUR TISE IS LINEAR

SIEGERT STATES



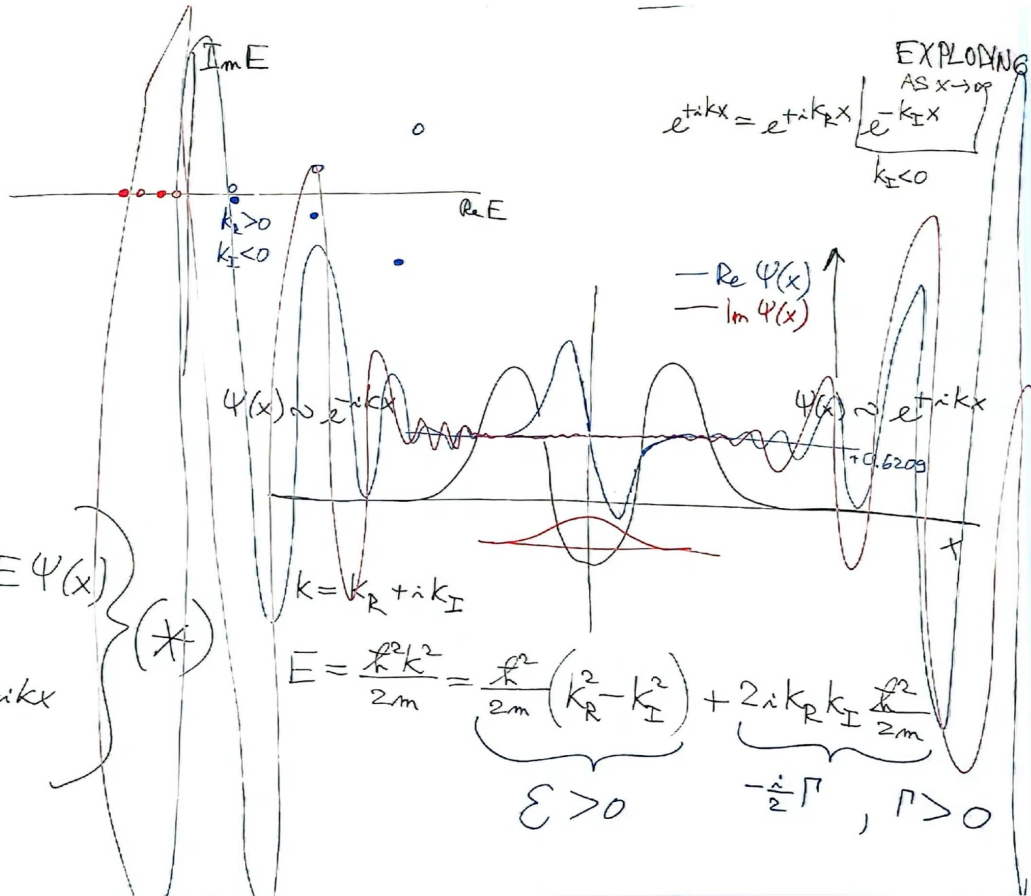
TISE

$$\left(-\frac{\hbar^2}{2m} \partial_{xx} + V(x)\right) \psi(x) = E \psi(x)$$

SIEGERT BOUNDARY CONDITIONS

$$\psi(x \rightarrow \pm\infty) = C_{\pm} e^{\pm i k x}$$

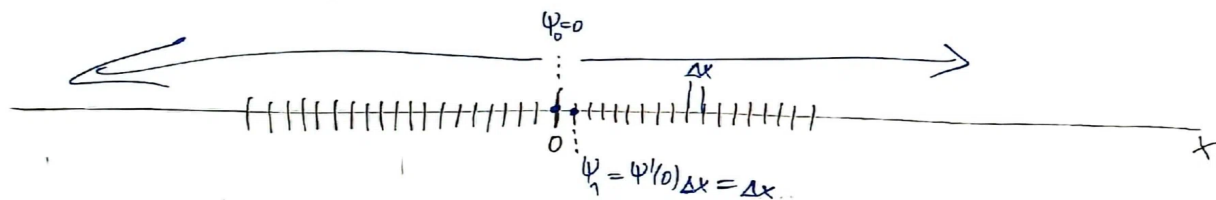
(*)



WE KNOW $E = \epsilon - \frac{i}{2}\Gamma = +0.6209 - \dots i$ 10^{-5}

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) + V(x) \psi(x) = E \psi(x)$$

IF $\psi(x)$ IS A SOLUTION,
THEN $\lambda \psi(x)$ IS A SOLUTION TOO



LOOKING FOR ANTI-SYMMETRIC SOLUTION: $\psi(0) = 0$

SYMMETRIC $\psi(0) \neq 0$ CHOOSE $\psi'(0) = 1$
 $\psi(0) = 1$
 $\psi'(0) = 0$

ANALYTIC CONTINUATION

$$f(x) = x^3 \sin x$$

$$\begin{aligned} f(z) &= z^3 & z \in \mathbb{C} \\ \sin z &= \frac{e^{+iz} - e^{-iz}}{2i} & z = x + iy \\ e^{iz} &= \sum_{n=0}^{\infty} \frac{1}{n!} z^n \end{aligned}$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (z-x_0)^n$$

#12 #13
FRIDAY AND TUESDAY: Q & A SESSION ABOUT THE PROJECT

LAST CHANCE TO ASK QUESTIONS! ✓

I WILL NOT ANSWER ANY QUESTIONS ABOUT THE PROJECT AFTER THE NEXT TUESDAY.

CONTINUE $\phi(x)$ ANALYTICALLY INTO THE COMPLEX PLANE:

$$\begin{array}{ccc} \psi(x) & \longrightarrow & \psi(z) \\ x \in \mathbb{R} & & z \in \mathbb{C} \end{array}$$

$\psi(xe^{i\theta})$ IS SQUARE

INTEGRABLE

FOR $\theta > \theta_{\text{CRITICAL}}$

COMPLEX ROTATION BY ANGLE θ

$$\theta > 0$$

FOR LARGE ENOUGH θ

$$e^{i\theta} = \cos\theta + i\sin\theta$$

 $x \rightarrow \infty$

$$e^{+ikx}e^{i\theta} = e^{+ik_R x}e^{i\theta} + |k_I| x e^{i\theta}$$

$$= \frac{e^{+ik_R x \cos \theta}}{e^{+ik_I x \cos \theta}} \frac{e^{+ik_I x \sin \theta}}{e^{+ik_R x \sin \theta}} = e^{(k_I \cos \theta - k_R \sin \theta)x}$$

OSCILLATING

$e^{F_\theta x}$ HOW DOES IT BEHAVE?

$$F_\theta = |k_I| \cos \theta - k_R \sin \theta$$

$\theta = 0$ $F_\theta = |k_I| > 0$ $e^{F_\theta x}$ EXPLODES AS $x \rightarrow \infty$

WE CHOOSE θ LARGE ENOUGH AS TO MAKE $F_\theta < 0$

$$F_{\theta_{\text{CRITICAL}}} = 0$$

$\theta > \theta_{\text{CRITICAL}}$ $F_\theta < 0$

$$|k_I| \cos \theta = k_R \sin \theta$$

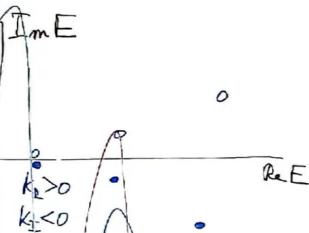
$$\tan \theta = \frac{|k_I|}{k_R}$$

TISE

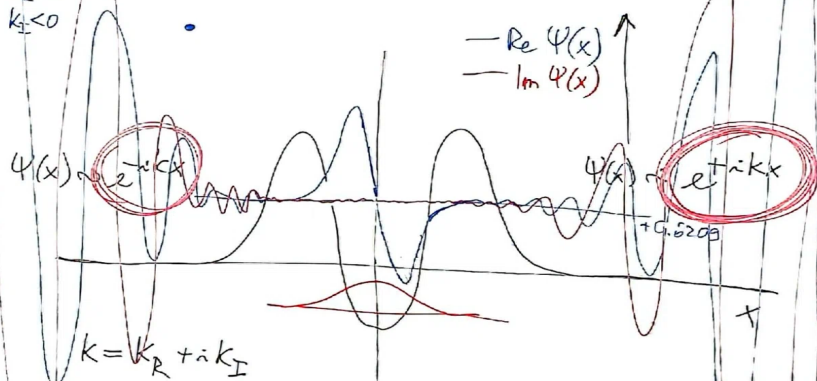
$$\left(-\frac{\hbar^2}{2m} \partial_{xx} + V(x) \right) \psi(x) = E \psi(x)$$

SIEGERT BOUNDARY CONDITIONS

$$\psi(x \rightarrow \pm \infty) = C_\pm e^{\pm i k x}$$



$$e^{+i k x} = e^{+i k_R x} e^{-k_I x} \quad \text{EXPLODING AS } x \rightarrow \infty \quad k_I < 0$$



$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m} (k_R^2 - k_I^2) + 2i k_R k_I \frac{\hbar^2}{2m}$$

$\mathcal{E} > 0$ $-\frac{i}{2}\Gamma, \Gamma > 0$

$e^{F_\theta x}$ HOW DOES IT BEHAVE?

$$F_\theta = |k_I| \cos \theta - k_R \sin \theta$$

$\theta = 0$ $F_\theta = |k_I| > 0$ $e^{F_\theta x}$ EXPLODES AS $x \rightarrow \infty$

WE CHOOSE θ LARGE ENOUGH AS TO MAKE $F_\theta < 0$

$$F_{\theta_{\text{CRITICAL}}} = 0$$

$\theta > \theta_{\text{CRITICAL}}$ $F_\theta < 0$

$$|k_I| \cos \theta = k_R \sin \theta$$

$$\tan \theta = \frac{|k_I|}{k_R}$$

TISE

$$\left(-\frac{\hbar^2}{2m} \partial_{xx} + V(x) \right) \psi(x) = E \psi(x)$$

SIEGERT BOUNDARY CONDITIONS

$$\psi(x \rightarrow \pm \infty) = C_\pm e^{\pm i k x}$$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dz^2} \psi(z) + V(z) \psi(z) = E \psi(z)$$

$$V(z) = (0.5z^2 - 0.8) e^{-0.1z^2}$$

$$z = x e^{i\theta}$$

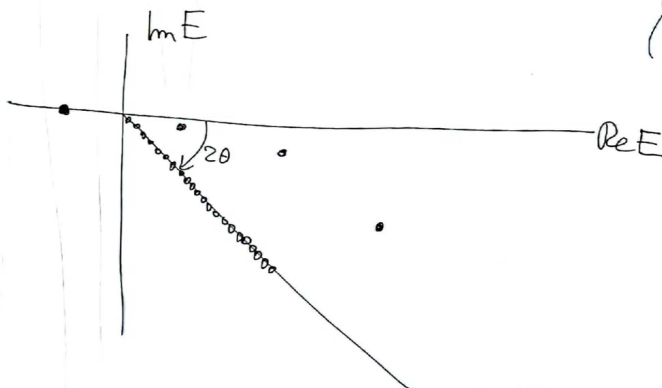
$$-\frac{\hbar^2}{2m} \frac{d^2}{d(xe^{i\theta})^2} \psi(xe^{i\theta}) + V(xe^{i\theta}) \psi(xe^{i\theta}) = E \psi(xe^{i\theta})$$

COMPLEX SCALED TISE

$$-\frac{\hbar^2}{2m} e^{-2i\theta} \frac{d^2}{dx^2} \psi_\theta(x) + V(xe^{i\theta}) \psi_\theta(x) = E \psi_\theta(x)$$

$$\psi_\theta(x) = \psi(xe^{i\theta})$$

INDEPENDENT OF θ
($\theta > \theta_{\text{CRITICAL}}$)



CONTINUE $\psi(x)$ ANALYTICALLY INTO

$\text{Im } z = y$

$\psi(xe^{i\theta})$
INTEGRABLE
FOR $\theta > \theta_0$

TWO MORE REMARKS:

1) PHYSICAL MEANING OF RESONANCES

$t=0$ INITIAL SQUARE INTEGRABLE
WAVEPACKET

$$\psi_0(x) = \psi(0, x)$$

$$i\hbar \partial_t \psi(t, x) = \hat{H} \psi(t, x)$$

IN THE INTERACTION REGION:

$$\psi(t, x) = \psi_{\text{RES}}(x) e^{-\frac{i}{\hbar}(\mathcal{E} - \frac{\Gamma}{2})t}$$

$$= \psi_{\text{RES}}(x) e^{-\frac{i}{\hbar}\mathcal{E}t} e^{-\frac{1}{2\hbar}\Gamma t}$$

$$|\psi(t, x)|^2 = |\psi_{\text{RES}}(x)|^2 e^{-\frac{\Gamma}{\hbar}t}$$

RESONANCE
WAVEFUNCTION
 $\psi_{\text{RES}}(x)$

$$\mathcal{E} - \frac{i}{2}\Gamma$$

